EEE115B *Digital Circuits*

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Sequential Logic

Programmable Logic

Random Access Memory (RAM)

Read-only Memory (ROM)

Programmable Logic Device

A programmable logic device (PLD) is one that does not initially have a fixed-logic function but that can be programmed to implement just about any logic design

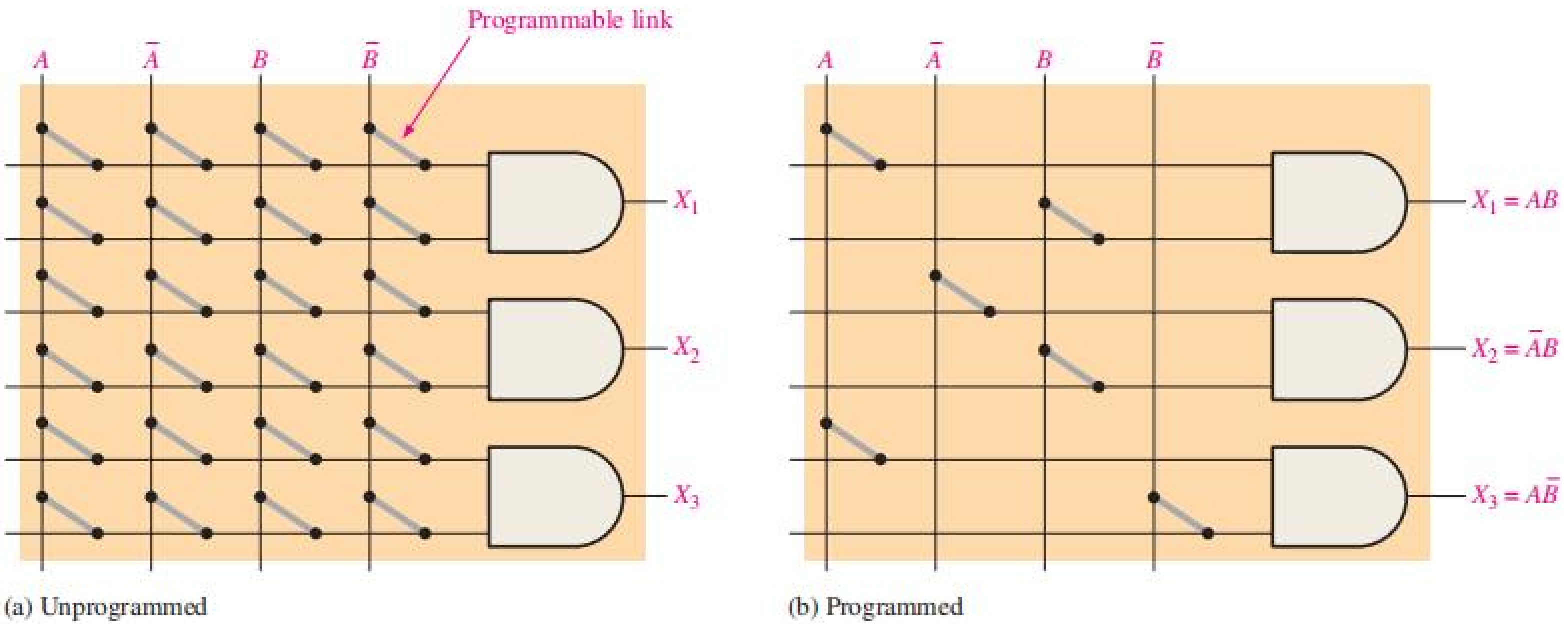
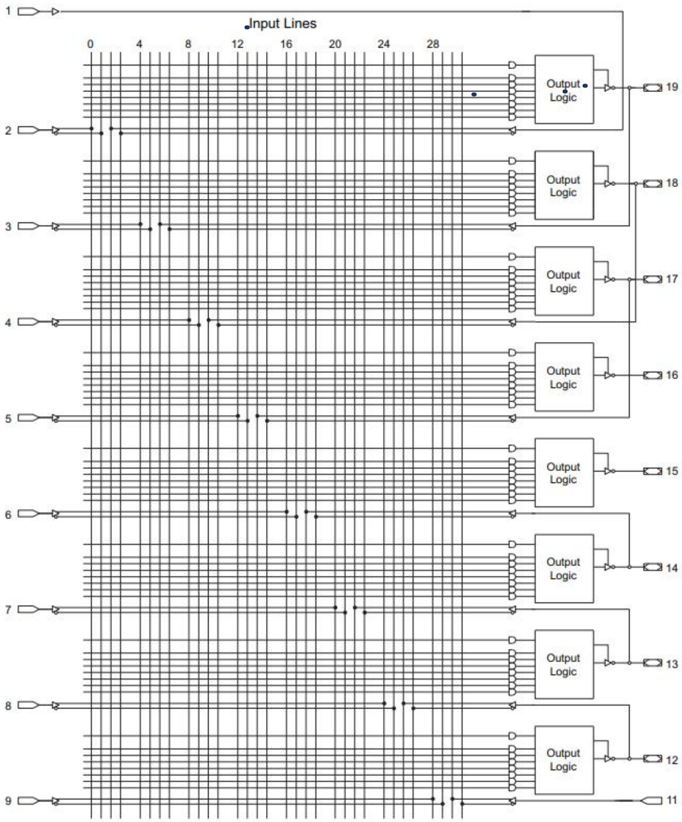


FIGURE 3–49 Concept of a programmable AND array.

Programmable Logic Device

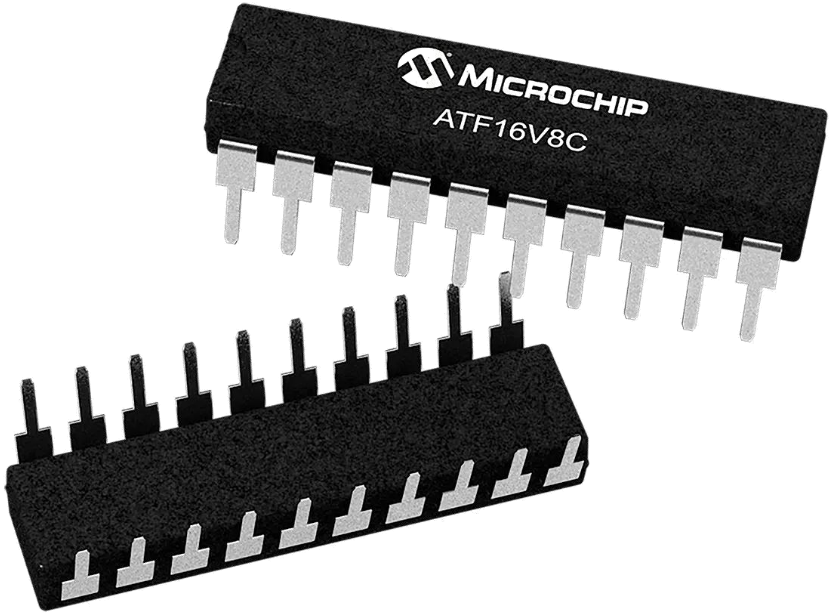
Figure 10-3. Simple Mode Logic Diagram



Note: 1. Input not available if power-down mode is enabled.

16 ATF16V8C [DATASHEET]
Atmel-0425I-PLD-ATF16V8C-Datasheet_032014

Atmel



Some Questions

Assume a PLD device:

- How to turn on/off these switches?
- Will logic lost after power off?

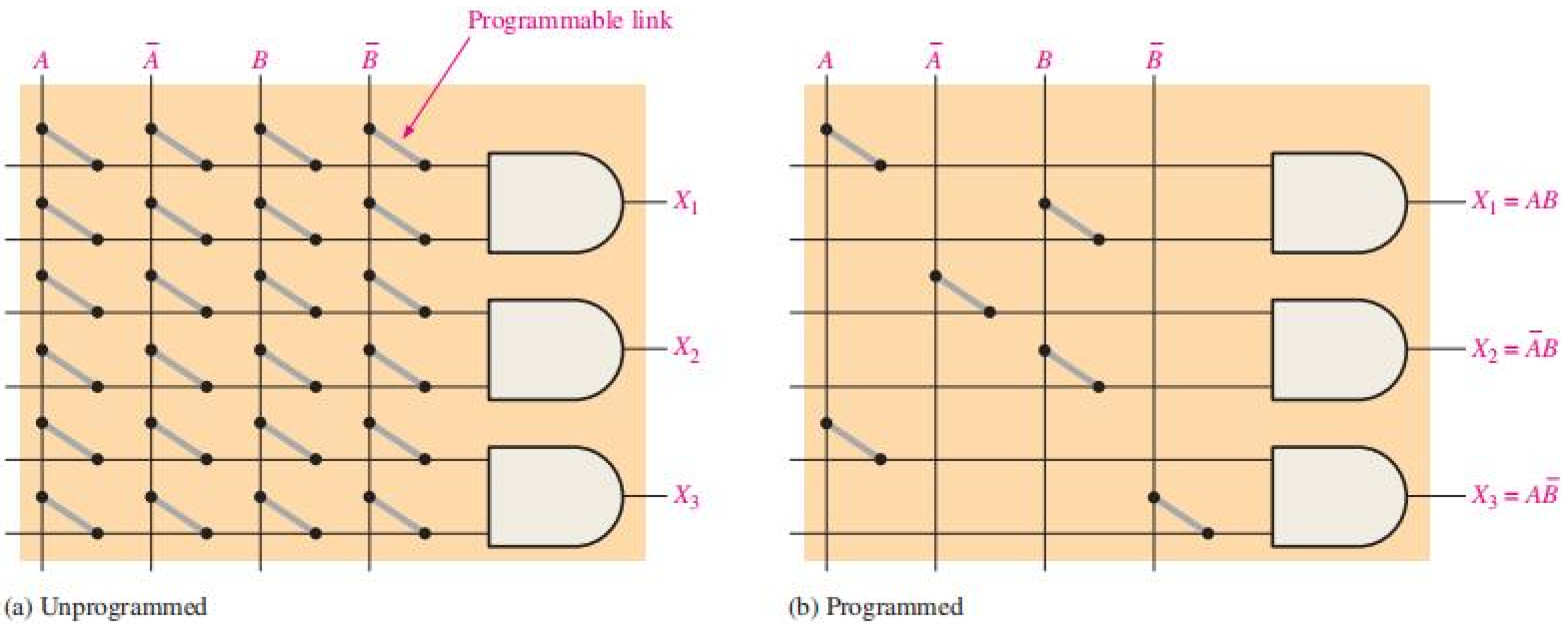
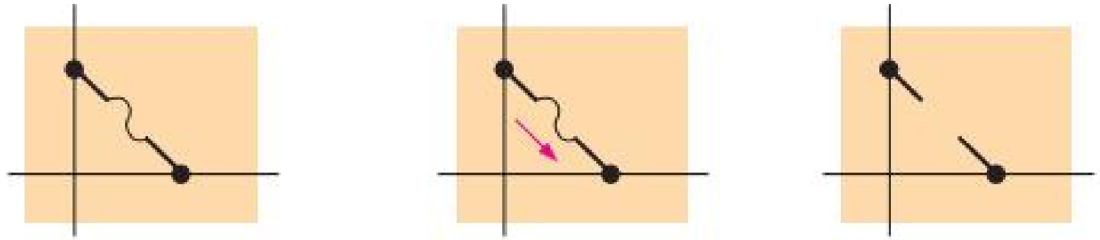


FIGURE 3–49 Concept of a programmable AND array.

Fuse Technology

Fuse Technology: One-time programmable (OTP)

To program a device, the selected fuses are opened by passing a current through them to “blow” the fuse and break the connection.



(a) Fuse intact before programming

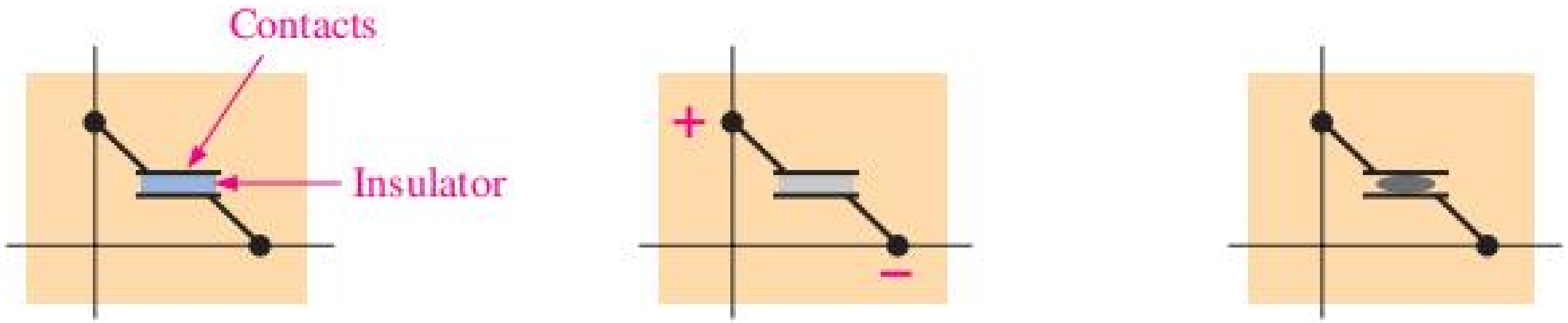
(b) Programming current

(c) Fuse open after programming

Antifuse Technology

Antifuse Technology: One-time programmable (OTP)

To program a device, sufficient voltage is needed to break down the insulator.



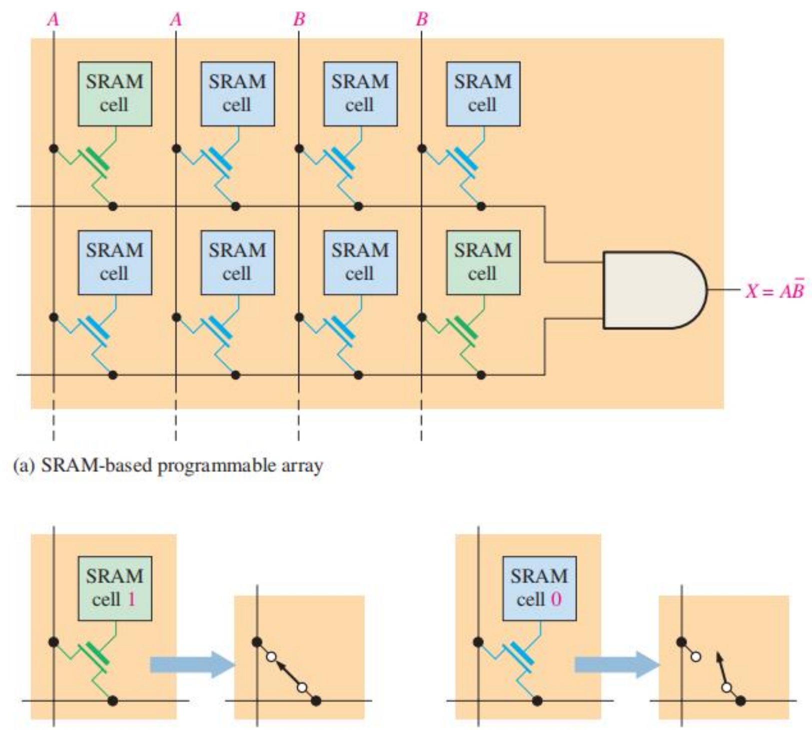
(a) Antifuse is open before programming.

(b) Programming voltage breaks down insulation layer to create contact.

(c) Antifuse is effectively shorted after programming.

Reprogrammable PLD (Volatile)

- A SRAM-type memory cell is used to turn a transistor on or off to connect or disconnect rows and columns.
- This means that a SRAM cell does not retain data when power is turned off.



(a) SRAM-based programmable array

(b) Transistor *on*

(c) Transistor *off*

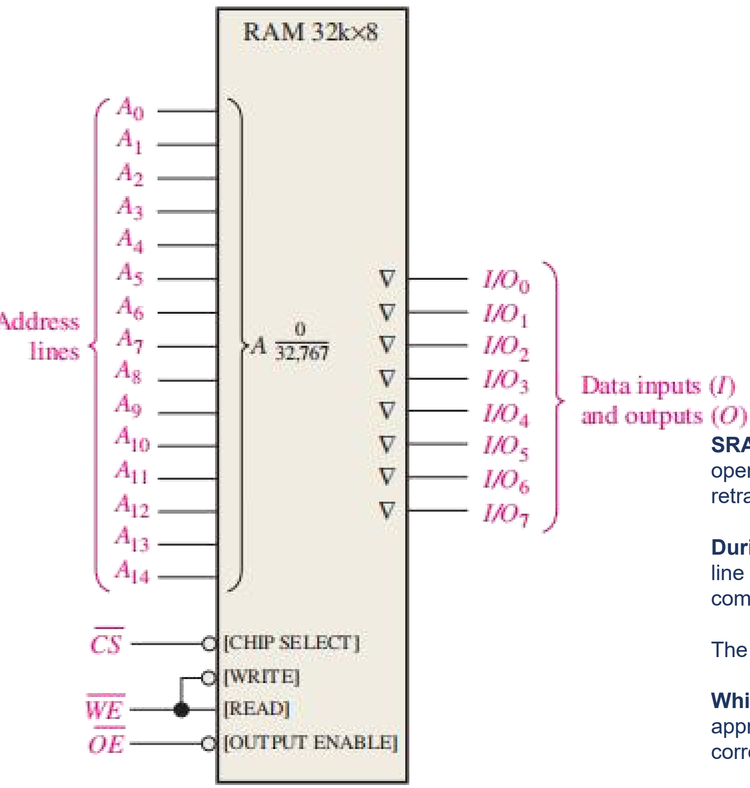
FIGURE 3–54 Concept of an AND array with SRAM technology.

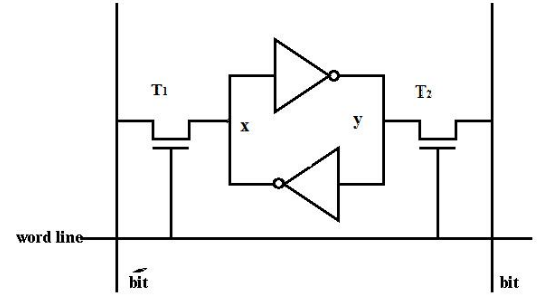
RAM & ROM

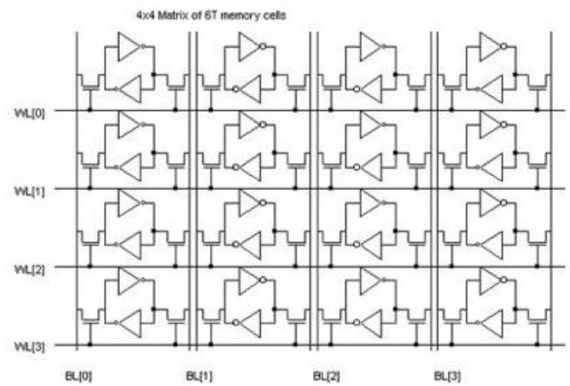
- RAM: Random access memory, fast, loses data when powered off.
  - SRAM: very fast, use transistors to store data bits. expensive to implement.
  - DRAM: slower than SRAM. use capacitor array.
- ROM: read-only memory, keeping its data even without power, slower than RAM.

Memory: Static RAM (SRAM)

RAM Size?
How to write?
How to read?







SRAM cell is make up of 2 inverters are cross-connected to form a latch (6 transistors). T1 and T2 can switch opened or closed under control of word line.When word line is ground level, the transistors are turned off and the latch retrains its state.

During the Read Operation,Word line is activated to close switches T1 and T2 .If the cell is in state 1; the signal on b line is high and the signal on b’ line is low.The opposite is true if the cell is in state 0.Thus, b and b’ are always complements of each other’s.

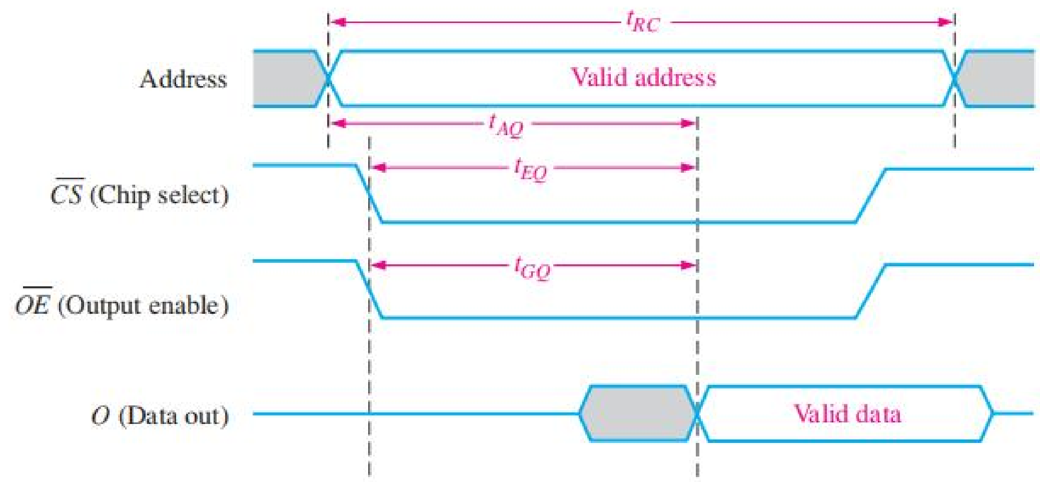
The sense/write circuit at the end of the two bit line monitors their state and sets the corresponding output accordingly.

While in Write Operation, the Sense/Write circuit drives bit lines b and b’, instead of sensing their state.It places the appropriate value on bit line b and its complement on b’ and activate the word line.This forces the cell into the corresponding state, which the cell retains when the word line is deactivated.

https://blackandwhitecomputer.blogspot.com/2012/03/reading-and-writing-operation-of-sram.html

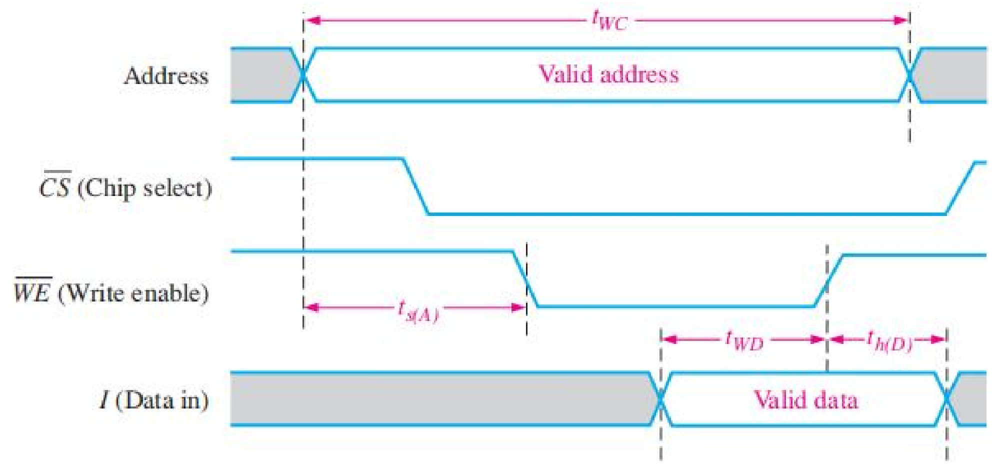
Memory: Static RAM (SRAM)

Read Logic



(a) Read cycle ($\overline{WE}$ HIGH)

Write Logic



(b) Write cycle ($\overline{WE}$ LOW)

Lock at WE rising edge

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Dynamic RAM (DRAM)

DRAM memory cell consists of one transistor and a capacitor and is simpler than the SRAM cell.

This allows much greater densities in DRAMs and results in greater bit capacities for a given chip area, although much slower access time.

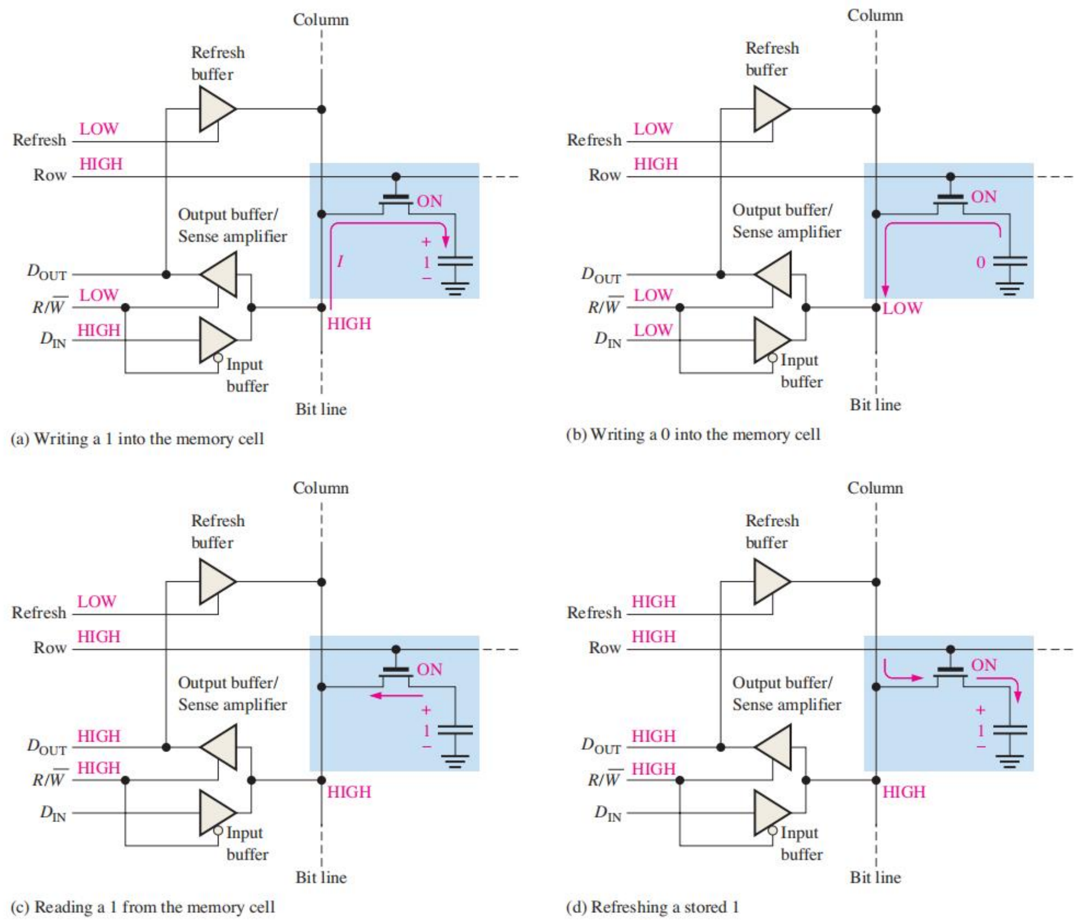

FIGURE 11–18 Basic operation of a DRAM cell.

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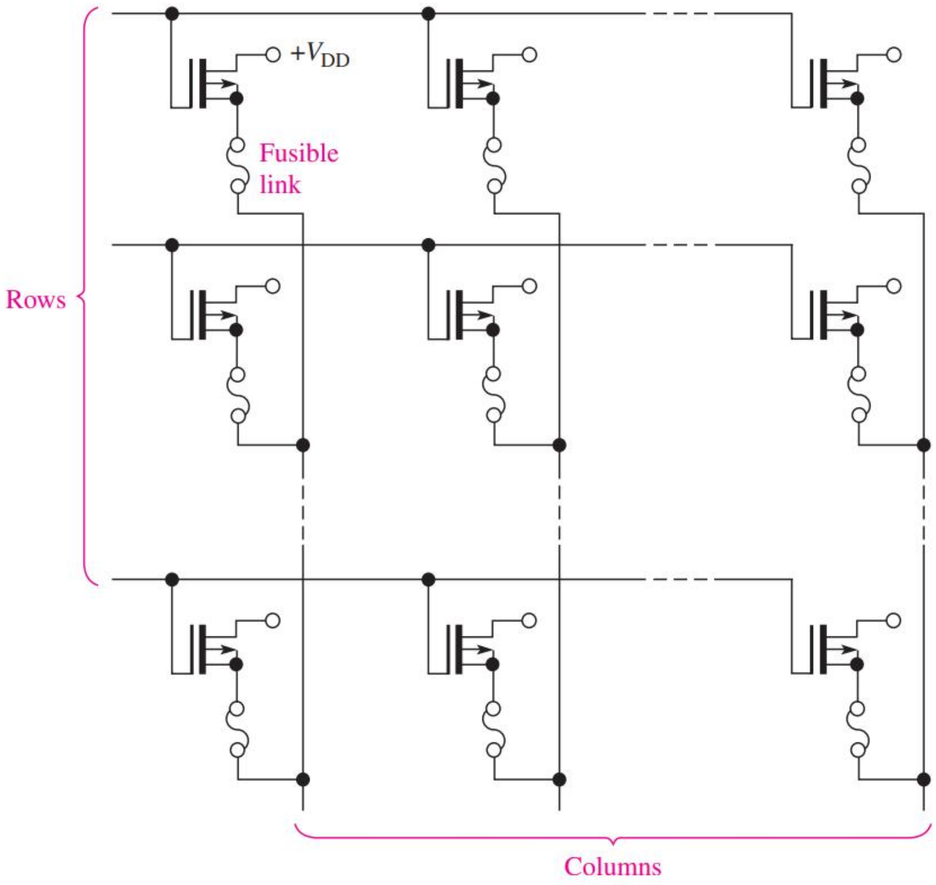
OTP ROM

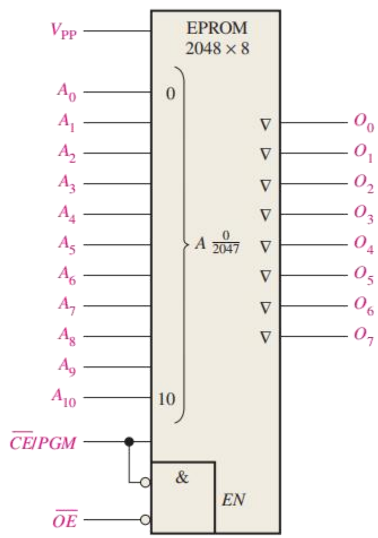
Write: use row and column to burn fuses.
ROW=1
COLUMN=0

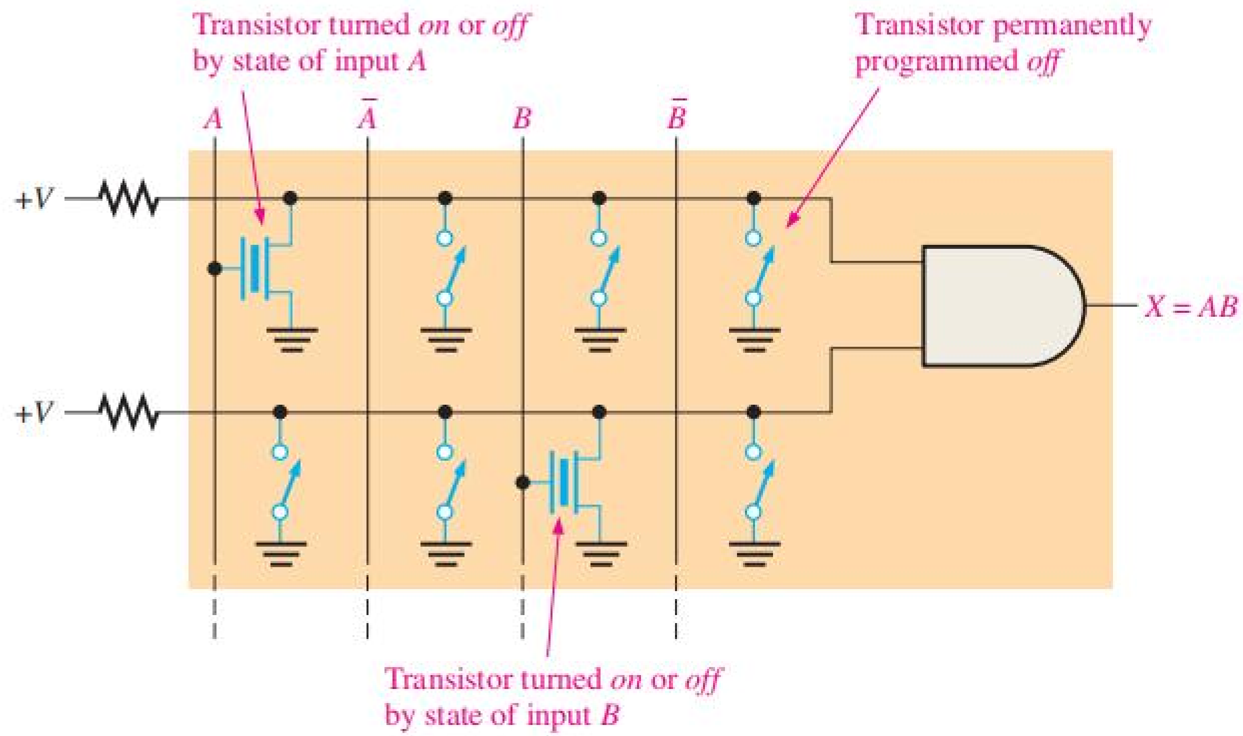
Read: use row and column to read fuse states.
ROW=1
COLUMN
=read with pull down



EPROM: Erasable Programmable Read-Only Memory

- Reprogrammable ROM is nicer to have.
- Floating-Gate transistor: store electrons.
- Can be erased with UV (ultraviolet) light and reprogrammed



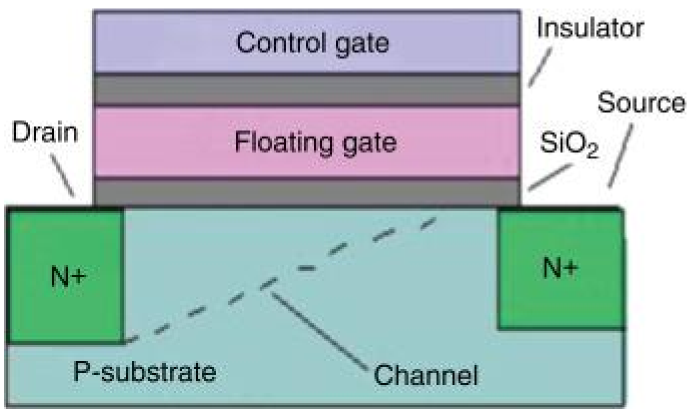


EEPROM Technology

- Electrically erasable programmable read-only memory technology
- EEPROM can be erased and reprogrammed electrically without the need for UV light or special fixtures

Flash Technology

Flash technology is based on a single transistor link and is both nonvolatile and reprogrammable. Flash elements are a type of EEPROM but are faster in writing, and result in higher density devices than the standard EEPROM link.

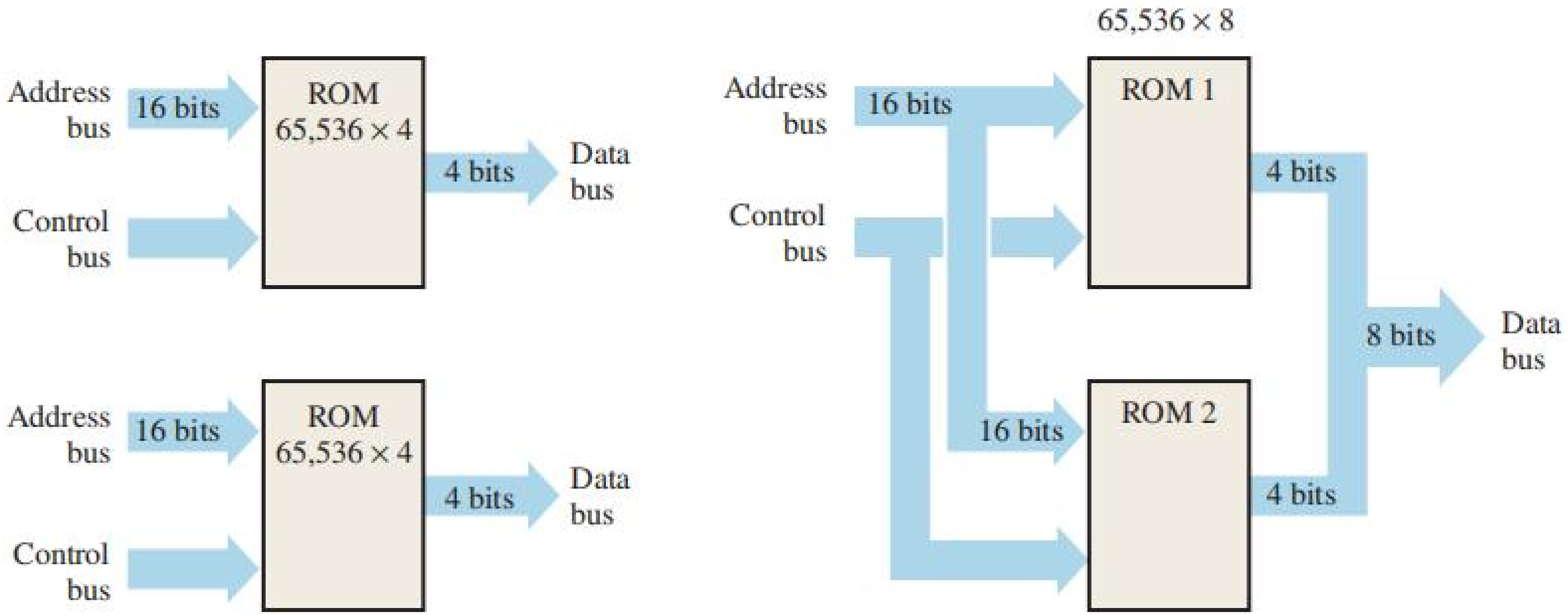


Electrons can be generated due to Fowler-Nordheim (FN) electron tunneling

https://www.sciencedirect.com/topics/engineering/floating-gate

Memory Expansion

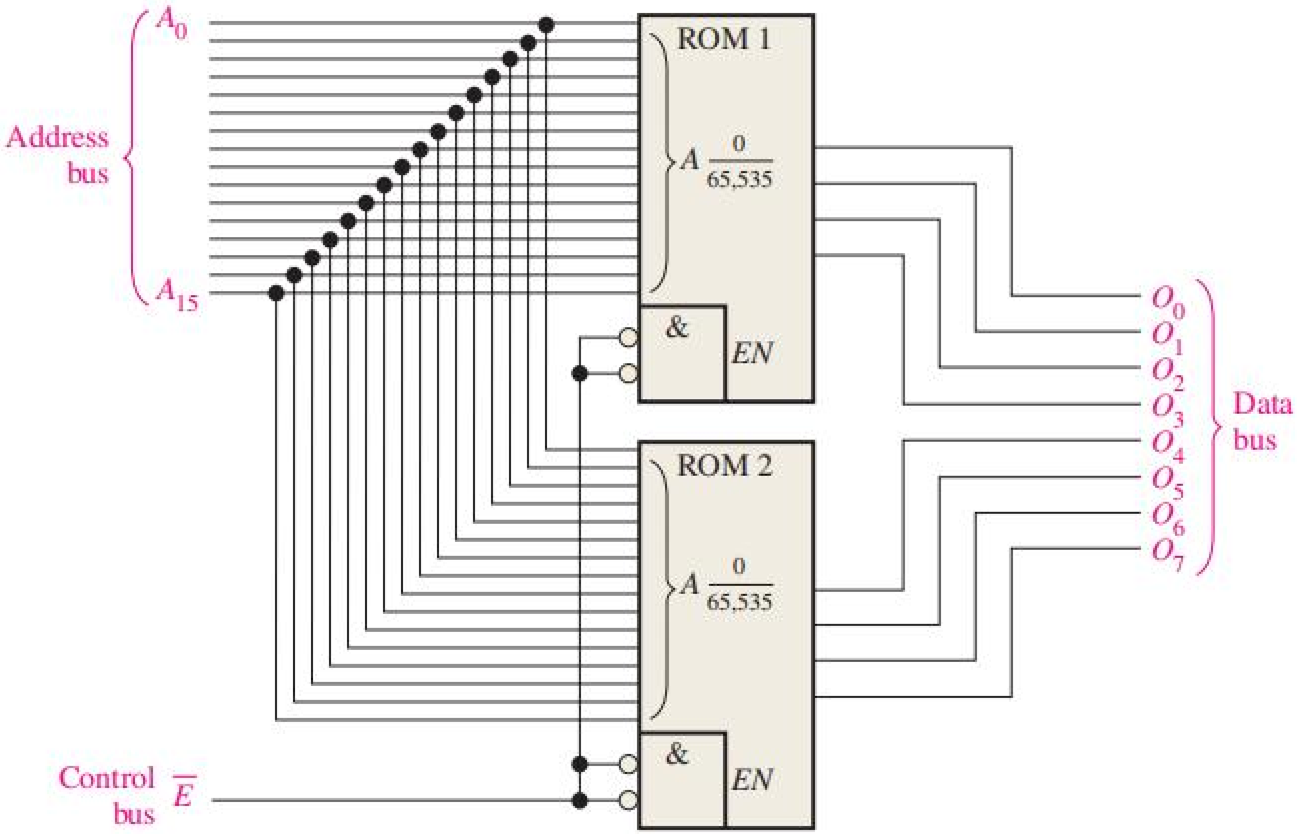
Word-Length Expansion



(a) Two separate 65,536 × 4 ROMs

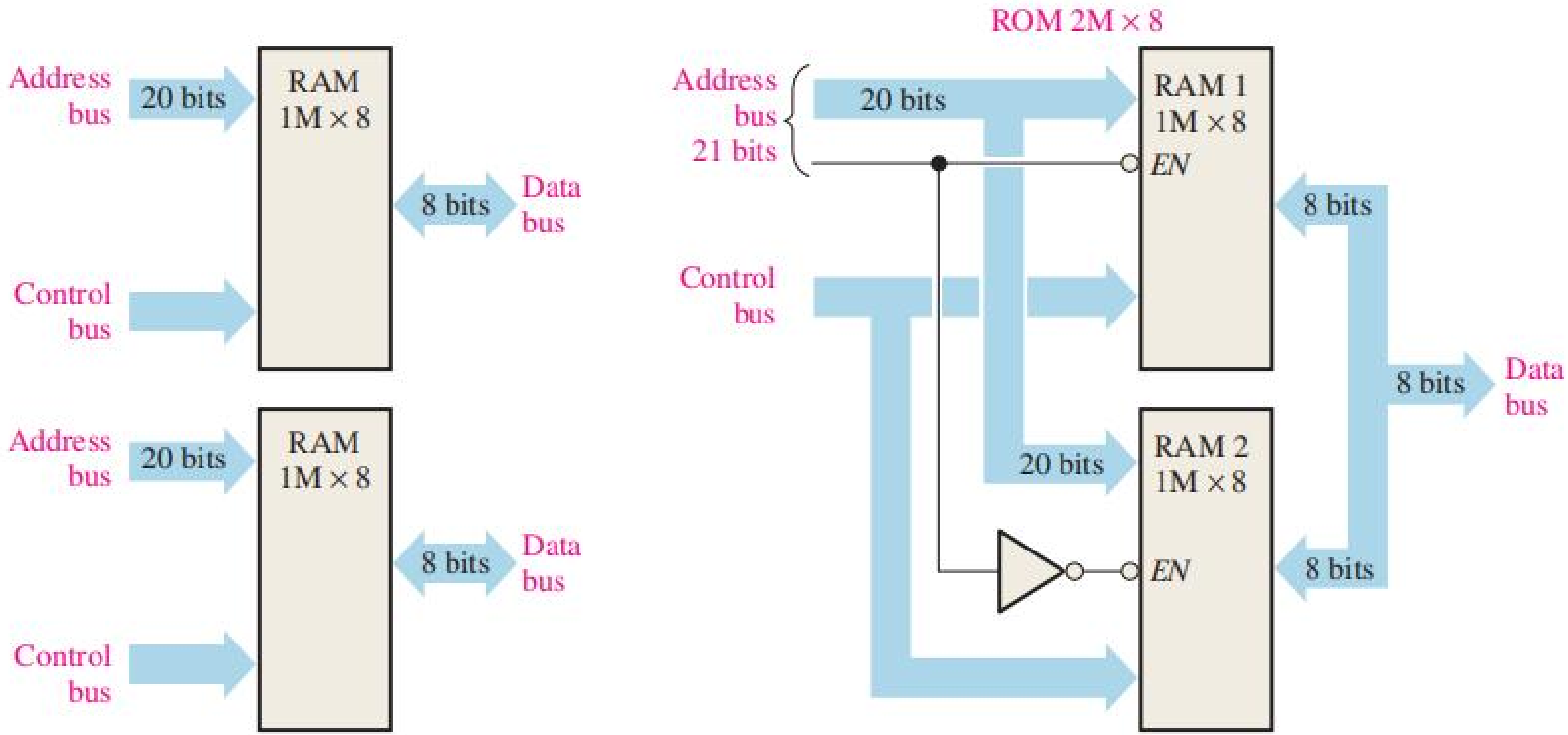
(b) One 65,536 × 8 ROM from two 65,536 × 4 ROMs

Word-Length Expansion



Word-Capacity Expansion

Connect two RAMs with 20 bit address lines, resulting in an interface with 21 bit address lines

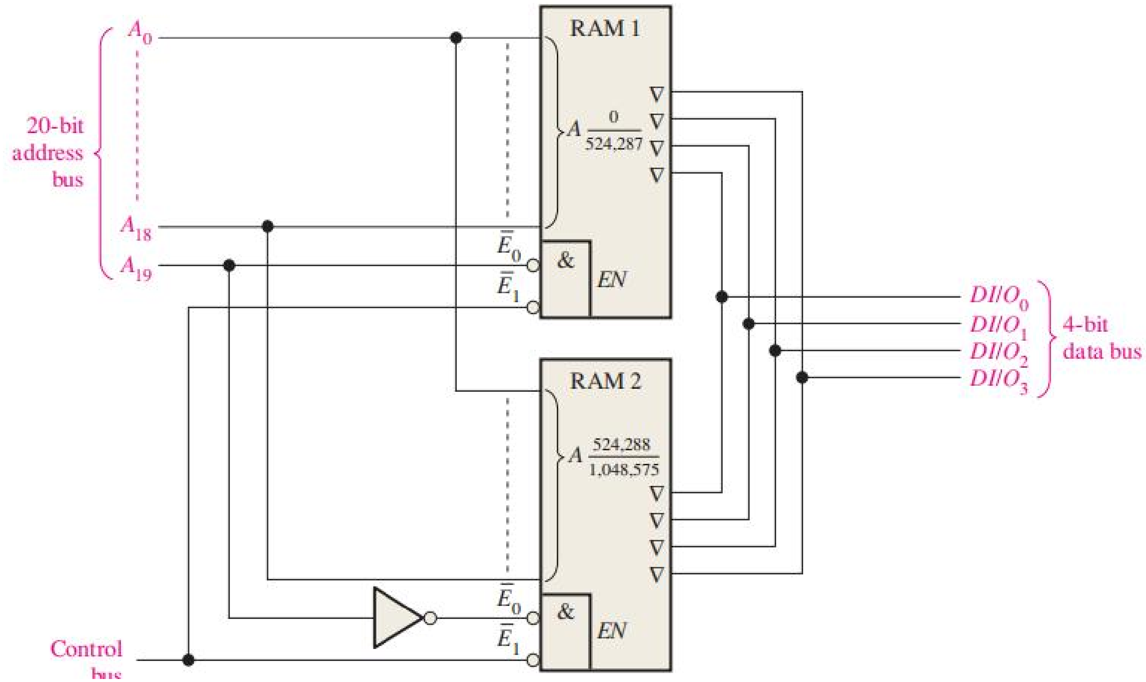


(a) Individual memories each store 1,048,576 8-bit words

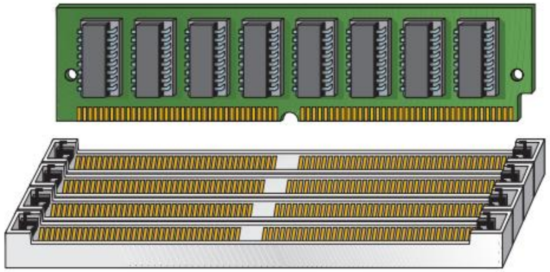
(b) Memories expanded to form a 2M × 8 RAM requiring a 21-bit address bus

Word-Capacity Expansion

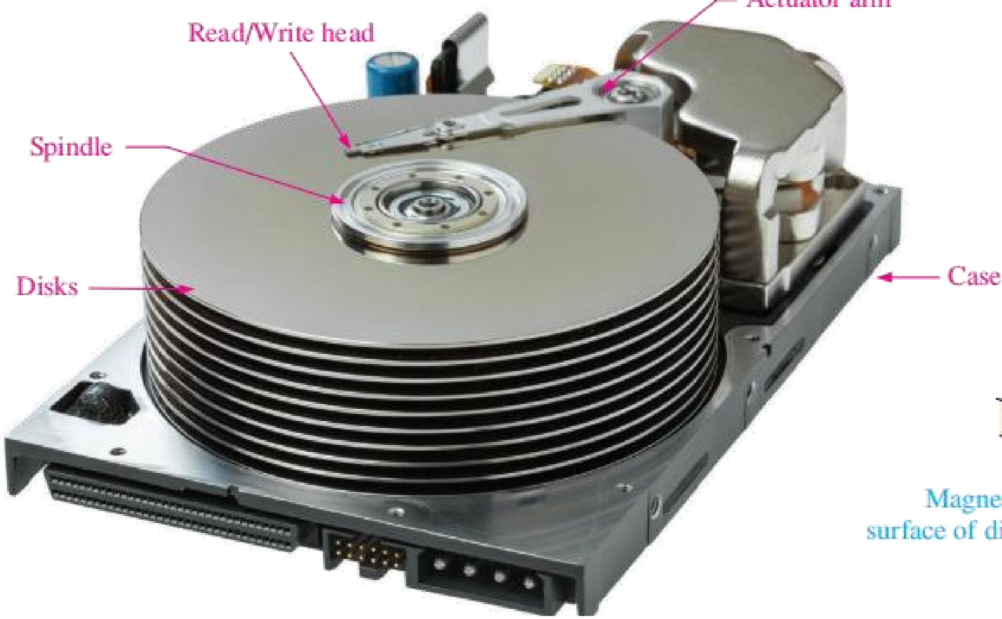
Connect two RAMs with 19 bit address lines, resulting in an interface with 20 bit address lines

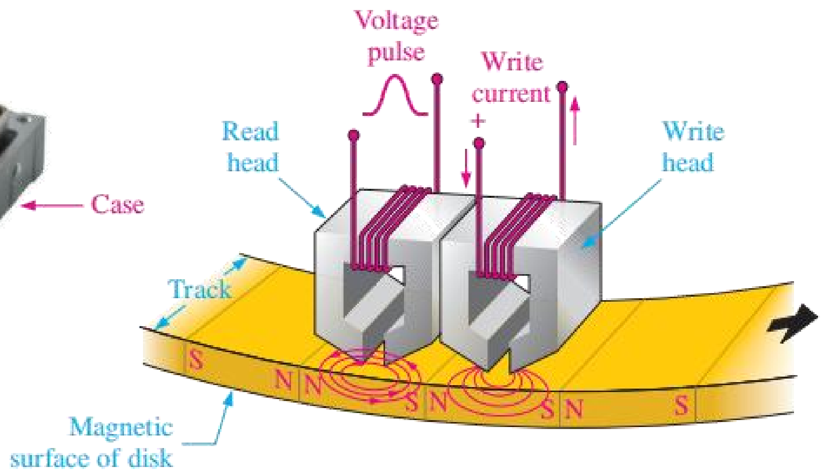


Pay attention to address.
Can you extend more?

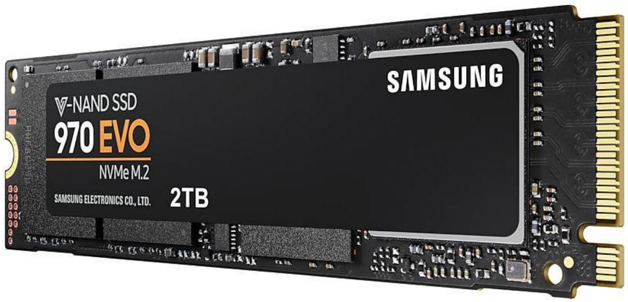


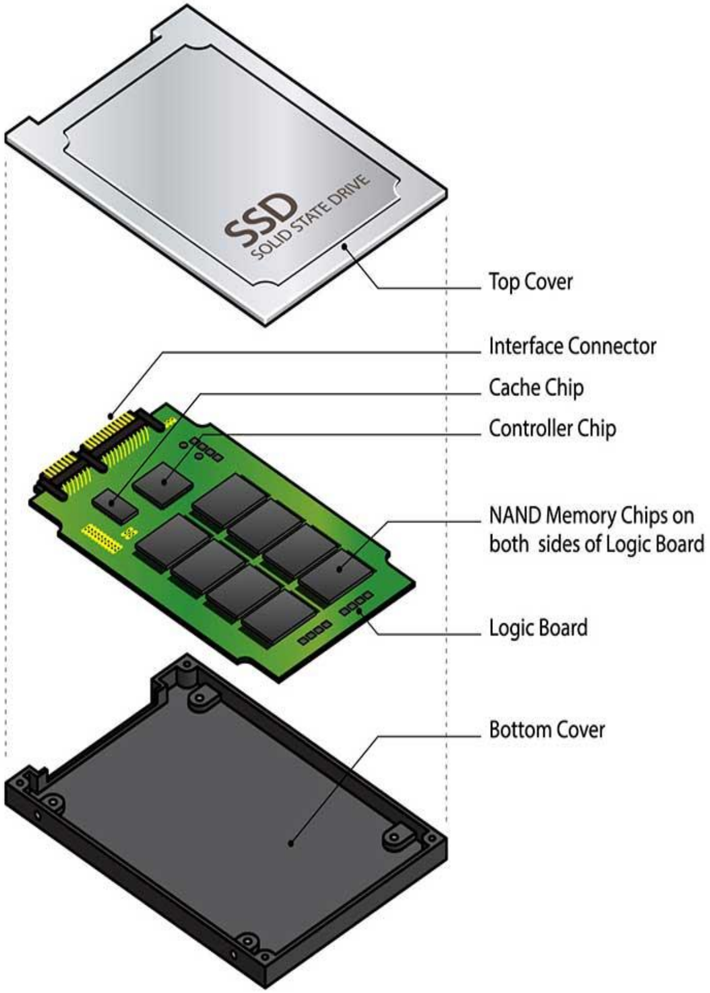
Magnetic Hard Disks





Solid State Hard Drives





CD-ROM

As the disk rotates, the narrow laser beam strikes the series of pits and lands of varying lengths, and a photodiode detects the difference in the reflected light. The result is a series of 1s and 0s corresponding to the configuration of pits and lands along the track

Prism